Predicted Student Misconceptions for Worksheet No. 2 (Bridge Modeling)

1. Contextual & Conceptual Misinterpretations

- **Misunderstanding the h-axis:** The problem defines h as the "distance **below** the base." Students often struggle with the idea of height being represented by negative values ($h \le 0$). They may instinctively flip the parabola upward because they associate a bridge arch with "going up" rather than being measured downward from a baseline.
- **Unit & Scale Realism:** Students might perform the algebra correctly but fail to attach the unit (meters) or may provide unrealistic dimensions because they lack a "sense of scale" for a real-world infrastructure project.

2. Task (a): Determining the Turning Point

- **Fractional Arithmetic Anxiety:** The coefficient $a = -\frac{1}{40}$ is a major hurdle.

  Common Error: When calculating $x = -\frac{b}{2a}$, students often fail to divide correctly by a fraction, resulting in $x = 0.05$ or $x = \frac{1}{20}$ instead of $x = 20$.
- **Substitution Mistakes:** When finding \$h\$ by substituting $x = 20$, students frequently make order-of-operation errors. They might square the entire term $\left(-\frac{1}{40} \cdot 20\right)^2$ instead of only squaring the x value (20^2).

3. Task (b): Defining the Domain (Possible values for x)

- **The "Half-Bridge" Error:** Many students find the axis of symmetry ($x = 20$) and stop there, concluding that x ranges from 0 to 20. They forget to apply the property of symmetry to find the full width of the arch (ending at $x = 40$).
- **Ignoring Physical Constraints:** Some may provide a purely mathematical domain ($x \in \mathbb{R}$) without restricting it to the interval $0 \le x \le 40$ mandated by the bridge's structure and the ground-level intercepts.

4. Task (c): Determining the Range of h

- **Maximum/Minimum Confusion:** Because $a < 0$, the vertex at $h = 0$ is the **maximum** value. Students often get confused because 10 is "bigger" than 0, mistakenly labeling $h = -10$ as the maximum or highest point.
- **Interval Notation Errors:** Students often write the range incorrectly as $[0, -10]$, failing to follow the standard mathematical convention of writing the smaller number first: $[-10, 0]$.

5. Task (d): Sketching the Graph

- **Lack of Boundary Awareness:** Students often draw a continuous, infinite parabola that extends below $h = -10$ or to the left of $x = 0$, ignoring the physical boundaries of the bridge supports.
- **Shape Inaccuracy:** Because $|a|$ is very small $\left(\frac{1}{40}\right)$, the parabola is quite "flat" (wide). Students used to drawing $y = x^2$ tend to draw the arch too narrow or "pointy," which does not align with the provided grid coordinates.
- **Grid Scaling Errors:** On the provided axes, the x-axis increments by 10 while the h-axis increments by 2. Students frequently plot points incorrectly because they do not pay close attention to the differing scales of the two axes.

ANSWER

Group:……..

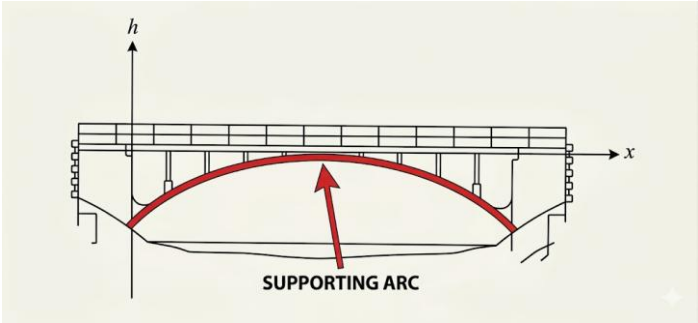
Class:……

Worksheet No. 2

Problem:

In 2026, a construction company is commissioned to design a modern transport bridge connecting Long Thanh International Airport to Ho Chi Minh City. The engineers propose a curved supporting arch modeled by the function:

$$h = -\frac{1}{40}x^2 + x - 10$$



where:

- *h (m)* is the distance below the base of the bridge.
- *x (m)* is the distance from the left side of the bridge.

You will become bridge engineers to analyse the model and prepare a short report by answering the following questions:

a) Determine the turning point of the graph.

The turning point of the graph $h = -\frac{1}{40}x^2 + x - 10$ has the coordinates $(20,0)$.

b) What are the possible values for x?

Since x is the distance from the left side of the bridge, $x \geq 0$.

The axis of symmetry is $x = 20$.

Thus, the possible values for x: $0 \leq x \leq 40$.

c) Determine the range of values of h.

Since h is the distance below the base of the bridge, $h \leq 0$.

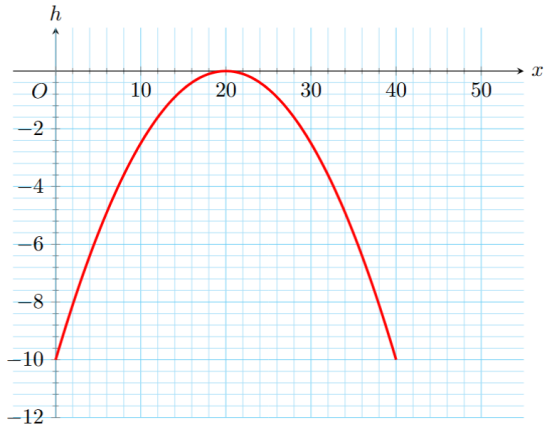
The y-intercept is -10.

Since $x \geq 0$, it follows that $h \geq -10$.

Thus, the range of values of h: $-10 \leq h \leq 0$.

d) Sketch the graph of the function $h = -\frac{1}{40}x^2 + x - 10$ within the possible values for x and h.

The graph of the function within the possible values for x and h is:


h
x
O
10
20
30
40
50
−2
−4
−6
−8
−10
−12